

ARASH JALIL KHABBAZI

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SUMMARY

I'm a PhD candidate in Mechanical Engineering at Purdue University, working on modeling, control, optimization, and machine learning for building energy systems. My research builds practical automation for efficient, occupant- and grid-interactive buildings, focusing on predictive control and reinforcement learning. I develop and demonstrate scalable data-driven control algorithms in real buildings, and I also work on power control systems in homes and on demand flexibility in commercial buildings. My work combines rigorous methods with real-world testing to make control systems more reliable and easier to deploy at scale.

EDUCATION

Purdue University, West Lafayette, IN PhD in Mechanical Engineering Concentration in Computational Science and Engineering	08/2023 – Present GPA: 3.9/4.0
University of British Columbia, BC, Canada MS in Mechanical Engineering Thesis: “Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines”	05/2021 – 08/2023 GPA: 94% (4.0/4.0)
University of Tabriz, EA, Iran BS in Mechanical Engineering	08/2016 – 07/2020 GPA: 19.12/20 (4.0/4.0) – Rank: 1/155+

SELECT AWARDS AND RECOGNITION

ASHRAE Graduate Student Grant-in-Aid Award	2025
Best PhD Forum Presentation Runner-up Award, ACM BuildSys	2025
ACM SIGEnergy Student Travel Grant, ACM BuildSys	2025
Best Paper Award, CSME International Congress	2023
Best Presentation Award, CSME International Congress (Advanced Energy Symposium)	2022
Graduate Research Scholarship, University of British Columbia	2022
Graduate Dean’s Entrance Scholarship, University of British Columbia	2021

PUBLICATIONS

Journal Articles

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, G.P. Henze, and K.J. Kircher, “Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: A review.” *Applied Energy*, 2025. 🔗
1. **A.J. Khabbazi**, M. Zabihi, R. Li, M. Hill, V. Chou, and J. Quinn, “Mixing hydrogen into natural gas distribution pipeline system through Tee junctions.” *International Journal of Hydrogen Energy*, 2024. 🔗

Preprints

1. L.D. Reyes Premer, **A.J. Khabbazi**, and K.J. Kircher, “Indirect Data-Driven Predictive Control and the State-Space Predictor.” *arXiv:2602.10936*, 2026. *Under review*.

Conference Papers

6. W.G. Dierking, **A.J. Khabbazi**, L.D. Reyes Premer, and K.J. Kircher, “Scalable Supervisory HVAC Control for Linear Objectives.” *65th IEEE Conference on Decision and Control (CDC)*, 2026.
5. L.D. Reyes Premer, E.N. Pergantis, **A.J. Khabbazi**, F. Wu, and K.J. Kircher, “Evaluating Power Control Strategies for Avoiding Residential Panel Upgrades in Cold Climates.” *ACEEE Summer Study on Energy Efficiency in Buildings*, 2026.
4. **A.J. Khabbazi** and K.J. Kircher, “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.” *American Control Conference (ACC)*, 2026.
3. E.N. Pergantis, L.D. Reyes Premer, **A.J. Khabbazi**, Priyadarshan, F. Wu, D. Ziviani, and K.J. Kircher, “Active current limiting control of residential appliances for breaker panel protection across the US: a preliminary study.” *CISBAT*, 2025.

2. **A.J. Khabbazi**, E.N. Pergantis, L.D. Reyes Premer, A.H. Lee, J. Ma, H. Liu, G.P. Henze, and K.J. Kircher, “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference, 2024*.
1. **A.J. Khabbazi**, M. Zabihi, R. Li, V. Chou, and J. Quinn, “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *Canadian Society for Mechanical Engineering (CSME) International Congress, 2023. Best Paper Award*.

Abstracts

3. **A.J. Khabbazi**, “PhD Forum Abstract: Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2025. Best PhD Forum Presentation Runner-up Award*.
2. **A.J. Khabbazi**, R. Li, and J. Quinn, “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *Canadian Society for Mechanical Engineering (CSME) International Congress, 2022. Best Presentation Award*.
1. **A.J. Khabbazi**, R. Li, J. Quinn, and M. Hoorfar, “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII), 2021*.

TALKS

8. “Small HVAC Control Demonstrations in Larger Buildings Often Overestimate Savings.” *American Control Conference (ACC), New Orleans, LA, 2026*.
7. “Field Demonstration of Advanced Commercial HVAC Control for Scalable Deployment.” *PhD Forum, 12th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), Golden, CO, 2025*.
6. “What Have We Learned From Field Demonstrations of Advanced Commercial HVAC Control?” *8th International High Performance Buildings Conference, Purdue University, West Lafayette, IN, 2024*.
5. “Field Demonstrations of Advanced Commercial HVAC Control.” *Intelligent Building Operations Workshop, Purdue University, West Lafayette, IN, 2024*.
4. “Mixing Gaseous Hydrogen into Natural Gas Distribution Pipelines.” *Herrick Graduate Seminar, Purdue University, West Lafayette, IN, 2023*.
3. “Blending of Hydrogen into a Natural Gas Distribution Pipeline in British Columbia through a Tee Junction for Reducing GHG Emissions.” *CSME International Congress, Université de Sherbrooke, QC, Canada, 2023*.
2. “Green Hydrogen Supply to Urban Infrastructure and Buildings through Blending into the Existing Grid.” *CSME International Congress, University of Alberta, AB, Canada, 2022*.
1. “The Blending and Transmission of Hydrogen and Natural Gas in Transmission and Distribution Pipelines.” *13th International Green Energy Conference (IGEC-XIII), 2021*.

RESEARCH EXPERIENCE

- Purdue University**, Graduate Research Assistant, Kircher Research Group, *West Lafayette, IN* 08/2023–Present
- Building supervisory control software for a live field demonstration of data-driven HVAC control in a campus building.
 - Co-developed a commissioning-free linear-programming supervisory HVAC controller (CDC’26).
 - Co-developed an indirect data-driven predictive control method (TPC) built on a state-space predictor (under review).
 - Derived a multi-zone model of how cross-zone heat transfer inflates measured HVAC savings (ACC’26; under review).
 - Co-developed active current-limiting control for home appliances that avoids panel upgrades (CISBAT’25, ACEEE’26).
 - Reviewed 100+ field demonstrations of MPC and reinforcement learning for HVAC (Applied Energy 2025).

INDUSTRY EXPERIENCE

- Purdue Energy & Utilities**, Graduate Researcher, Energy Systems, *West Lafayette, IN* 02/2026–Present
- Leading modeling and data pipelines for a pre-cooling pilot to shift demand off-peak on a 40,320-ton chilled-water system.
 - Quantified ~\$41k/yr savings and a statistically significant 8% lower approach from a two-plant air/dirt-separator retrofit.
 - Delivered a nine-year energy-use assessment of 21 renovations to Purdue leadership for the Giant Leaps Master Plan.
- Tesla, Inc.**, Applied Machine Learning Intern, Drive Unit, *Sparks, NV* 05/2025–08/2025

- Built MySQL pipelines, engineered features, and trained ML models (XGBoost, random forests, neural nets) with R^2 0.85.
- Helped deploy a thermal imaging system for drive-unit inspection, replacing destructive laboratory testing.
- Delivered ML-driven quality gains projected to cut testing costs by over \$1M/year; offered a three-month extension.

FortisBC, Graduate Researcher, Renewable Gas, *Vancouver, BC*

05/2021–08/2023

- Investigated hydrogen blending into natural gas distribution systems on an NSERC-funded UBC-FortisBC project.
- Performed CFD simulations and developed ideal- and real-gas mixture models in C. 
- Contributed to hydrogen research within the UBC H₂Lab, established through this collaboration. 
- Delivered research resulting in one journal paper, three conference papers, an MS thesis, and two awards.

TEACHING EXPERIENCE

Purdue University, Graduate Teaching Assistant (PhD)

- ME597: Distributed Energy Resources (DERs)  *Spring'26*
 - Supported 52 students on modeling and control of DERs such as batteries, solar PV, EVs, and HVAC.
 - Assisted with homework, coding modules, and office hours covering DER concepts.
 - Guided semester projects on DER topics, including building controls and data center cooling.

University of British Columbia, Graduate Teaching Assistant (MS)

- Engineering Analysis I *Fall'21, Spring'22, Fall'22*
- Fluid Mechanics II Lab *Spring'22, Spring'23*

LEADERSHIP & MENTORSHIP

Research Mentor, Purdue University

2024–Present

- Mentored three master's students and one undergraduate on research planning and technical writing.
- Collaborated on their research and co-authored the resulting publications.

Disability Resource Centre Invigilator, University of British Columbia

2021–2023

- Invigilated accommodated exams, supporting students with approved testing accommodations.

ACADEMIC SERVICE

Conference and workshop organizing:

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| July 2026 | Student Chair, Intelligent Building Operations (IBO) Workshop  <i>West Lafayette, IN</i> |
| July 2024 | Student Chair, Intelligent Building Operations (IBO) Workshop, <i>West Lafayette, IN</i> |

Society memberships: ASHRAE, ACM, IEEE, CSME

Reviewing: IEEE Transactions on Control Systems Technology; IEEE Conference on Decision and Control; Building and Environment; Journal of Building Performance Simulation; ACEEE Summer Study on Energy Efficiency in Buildings; Herrick Conferences

SELECT COURSES

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| • ECE570: Artificial Intelligence | • STAT511: Statistical Methods |
| • IE690: Reinforcement Learning & Control (RLC) | • ME597: Industrial Internet of Things (IIoT) |
| • AAE568: Applied Optimal Control & Estimation | • ME597: Distributed Energy Resources (DERs) |
| • AAE 561: Intro to Convex Optimization | • ME511: Analysis of Thermal Systems |

SKILLS

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| Languages: | Python, MATLAB, C, SQL |
| Optimization & ML: | CVXPY, Gurobi, TensorFlow, PyTorch, scikit-learn, XGBoost, Stable-Baselines3 |
| Simulation & Testbeds: | BOPTEST, Gymnasium, OCHRE |
| Systems & Tools: | InfluxDB, Grafana, Git, Linux, L ^A T _E X, Docker |